INCH-POUND

MIL-C-55302/55F
23 September 1992
SUPERSEDING
MIL-C-55302/55E
26 August 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PLUG, SOCKET CONTACTS, DECADE INCREMENTS 10 THRU 70 CONTACT POSITIONS AND INTERMEDIATE POSITIONS OF 14, 24, 44, 54 AND 26, 36, 56, 66 FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

(F)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation. MIL-C-55302

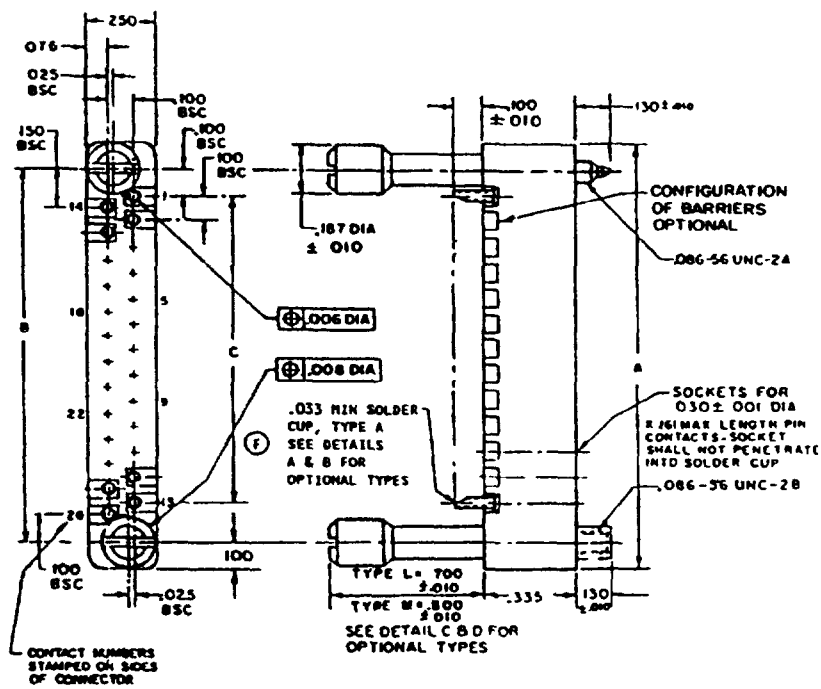
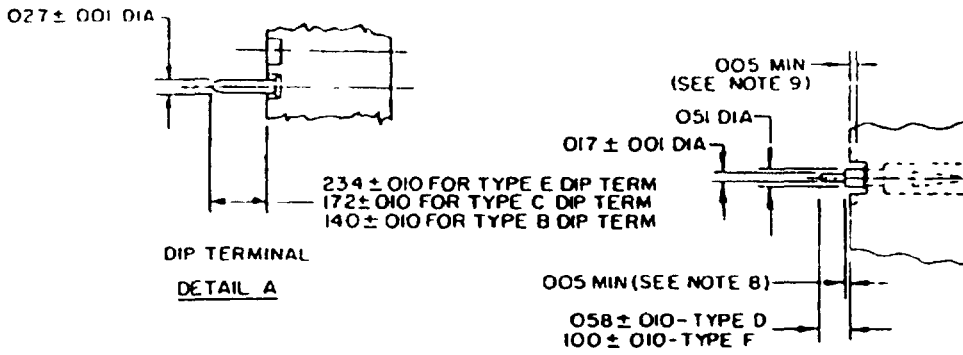


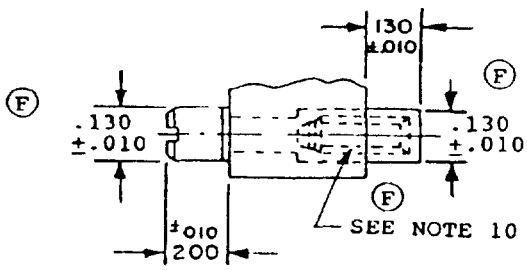
FIGURE 1 Connector, plug 100 (2.54 mm) spacing

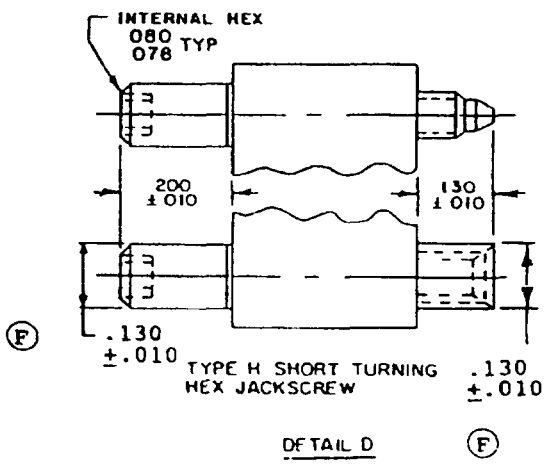
(F) denotes changes

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Number of contacts	Dimensions		
	A ± .010	BSC B	REF C
10	.850 (21.59)	.650 (16.51)	.400 (10.16)
14	1.050 (26.67)	.850 (21.59)	.600 (15.24)
20	1.350 (34.29)	1.150 (29.21)	.900 (22.86)
24	1.550 (39.37)	1.350 (34.29)	1.100 (27.94)
26	1.650 (41.91)	1.450 (36.83)	1.200 (30.48)
30	1.850 (46.99)	1.650 (41.91)	1.400 (35.56)
36	2.150 (54.61)	1.950 (49.53)	1.700 (43.18)
40	2.350 (59.69)	2.150 (54.61)	1.900 (48.26)
44	2.550 (64.77)	2.350 (59.69)	2.100 (53.34)
50	2.850 (72.39)	2.650 (67.31)	2.400 (60.96)
54	3.050 (77.47)	2.850 (72.39)	2.600 (66.04)
56	3.150 (80.01)	2.950 (74.93)	2.700 (68.58)
60	3.350 (85.09)	3.150 (80.01)	2.900 (73.66)
66	3.650 (92.71)	3.450 (87.63)	3.200 (81.28)
70	3.850 (97.79)	3.650 (92.71)	3.400 (86.36)

(F) FIGURE 1 Connector, plug .100 (2.54 mm) spacing - Continued

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Inches	mm	Inches	mm	Inches	mm
.001	0.03	.051	1.30	.171	4.09
005	0.13	.058	1 47	172	4.37
.006	0.15	075	1.91	.187	4.75
008	0.20	078	1.98	200	5.08
010	0.25	086	2.18	234	5.94
017	0 43	100	2 54	250	6 35
025	0 64	108	2 74	.335	8.51
027	0 69	130	3 30	500	12 70
.030	0 76	.140	3 56	.700	17.78
.035	0.89	.150	3 81		

NOTES

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm)
4. These connectors mate with connectors specified in MIL-C-55302/56.
5. Numbers indicating the first and last position in each row and every fourth contact position in between shall be marked on the side(s) of the connector. As an option to the above, numbers indicating every fourth cavity may be stamped on the side of the connector, with the exception that the number one contact shall be marked
6. Termination layout on .025 (0 64 mm) modular grid.
7. Metric equivalents are in parentheses
8. Conformal coating clearances shall be .005 (0.13 mm) minimum (not required for optional design).
9. Optional design conformal coating clearance
10. (F) Hole depth to female guide hardware is .282 min (7.16 mm) Full thread depth to female threaded hardware is .240 min (6 1 mm)

(F) FIGURE 1 Connector, plug .100 (2.54 mm) spacing - Continued.

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REQUIREMENTS

Dimensions and configuration See figure 1

Material Jackscrews and jacksockets shall be made of corrosion resisting stainless steel in accordance with ASTM-A581 or ASTM-A582 passivated in accordance with QQ-P-35 or shall be corrosion resistant copper nickel alloy (61 ±2% copper, 25 ±2% nickel, 2 5% max other, and the balance zinc) and shall have a brinell hardness of 145 to 175

Plating The contact plating shall be in accordance with MIL-C-55302 or gold in accordance with MIL-G-45204, type II, grade C, class 1, over nickel plating in accordance with QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold in accordance with MIL-G-45204, type II, grade C, class 00, or type III, grade A, class 00, over nickel plating in accordance with QQ-N-290, class 02, 50 to 150 microinches in the termination area

Contact identification See figure 1

Contact engagement and separation forces (No 22 contacts)
- Minimum separation: 0 5 ounce
- Maximum engagement: 12.0 ounces for standard force and 4.0 ounces for low force.
- Note Low force contacts meet the requirements of standard force contacts

Mating and unmating:
- Standard insertion force contacts The maximum mating force, in pounds, shall be the number of contacts multiplied by 0 56 and the withdrawal force, in pounds, shall be a minimum of 0 08 times the number of contacts and shall not exceed the measured insertion force

- Low insertion force contacts The maximum mating force, in pounds, shall be the number of contacts multiplied by 0 25 and the withdrawal force, in pounds, shall be a minimum of 0 04 times the number of contacts and shall not exceed the maximum insertion force

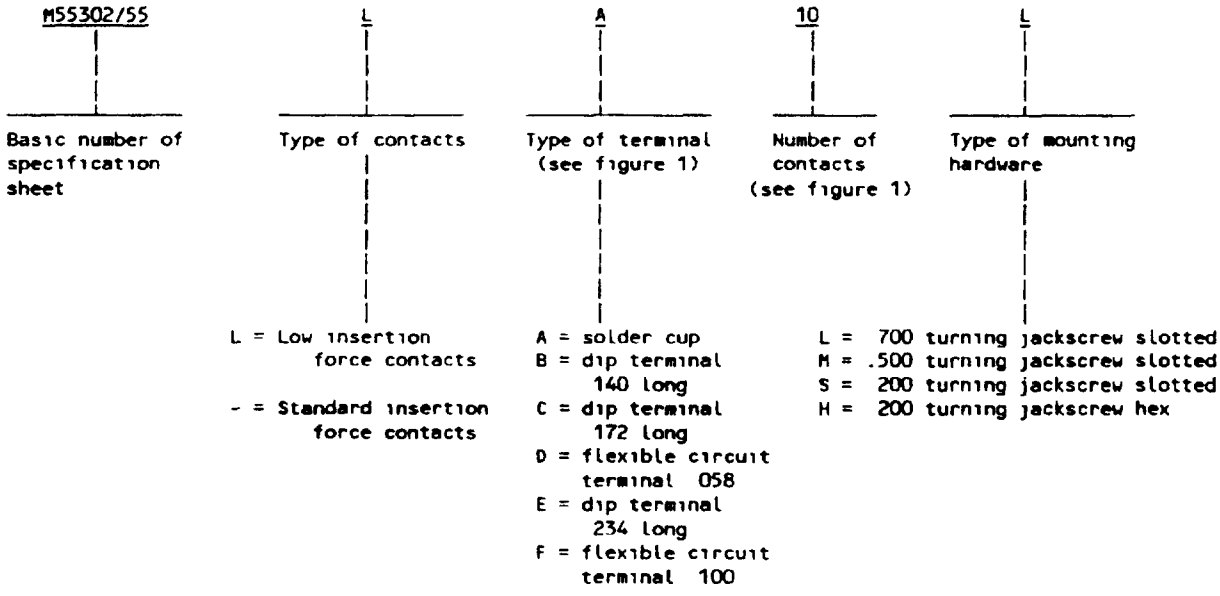
Contact resistance. The average contact resistance of all contacts measured shall not exceed 0 010 ohm, and no individual contact pair shall have a resistance exceeding 0 020 ohm

Dielectric withstanding voltage:
- Sea level: 1,000 volts rms.
- High altitude: 300 volts rms

Current rating 5.0 amperes maximum in accordance with contact.

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Example of part number



CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, TD
Air Force - 85, 99
DLA - ES

User activities
Army - AT, AV, ME
Navy - MC, OS
Air Force - 11, 19, 80

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3860-01)

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